

Class Work(28.07.26)

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QUESTIONS

Q1 
Void Function (No Return Value)
5 marks

Q2 
Void Function (No Return Value)
5 marks

Q3 
Void Function (No Return Value)
5 marks

Q4 
Fruitful Function (Returns One Value)
5 marks

Q5 
Fruitful Function (Returns One Value)
5 marks

Q6 
Fruitful Function (Returns One Value)

Q1 Void Function (No Return Value)

5 marks

Write a Python program to define a function that displays "Welcome to Python Programming".

Your Code

```
1 def welcome():  
2     print("Welcome to Python Programming")  
3  
4 welcome()
```

Input

stdin input

Output

Welcome to Python Programming

 Run Save

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Q6

Fruitful Function (Returns One Value)

Q2 Void Function (No Return Value)

5 marks

Write a function to print the multiplication table of a given number.

Your Code

```
1 def print_table(n):  
2     for i in range(1, 11):  
3         print(f"{n} x {i} = {n*i}")  
4  
5 num = int(input("Enter a number: "))  
6 print_table(num)
```

 Run Save

Input

5

Output

```
Enter a number: 5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30
```

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5 marks**Q6**

Fruitful Function (Returns One Value)

Q3 Void Function (No Return Value)

5 marks

Write a function to print all even numbers from 1 to N.

Your Code

```
1 def print_even_numbers(N):  
2     for i in range(1, N + 1):  
3         if i % 2 == 0:  
4             print(i)  
5  
6 N = int(input("Enter N: "))  
7 print_even_numbers(N)
```

 Run Save**Input**

15

Output

```
Enter N: 2  
4  
6  
8  
10  
12
```

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Fruitful Function (Returns One Value)

5 marks

**Q6**

Fruitful Function (Returns One Value)



Q4 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a string and returns the number of vowels in it.

Your Code

```
1 def count_vowels(s):  
2     vowels = "aeiouAEIOU"  
3     count = 0  
4     for ch in s:  
5         if ch in vowels:  
6             count += 1  
7     return count  
8  
9 s = input().strip()  
10 print(count_vowels(s))
```

 Run Save

Input

programming

Output

3

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5 marks

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Fruitful Function (Returns One Value)

5 marks

Q6

Fruitful Function (Returns One Value)

Q5 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a list of integers and returns the largest element.

Your Code

```
1 def largest_element(nums):  
2     return max(nums)  
3  
4  
5 if __name__ == "__main__":  
6     data = input().split()  
7     nums = list(map(int, data))  
8     print(largest_element(nums))
```

 Run Save

Input

3 1 9 7

Output

9

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Fruitful Function (Returns One Value)



Q6 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a number and returns whether it is prime or not.

Your Code

```
1 def is_prime(n):  
2     if n <= 1:  
3         return False  
4     if n == 2:  
5         return True  
6     if n % 2 == 0:  
7         return False  
8     i = 3  
9     while i * i <= n:  
10        if n % i == 0:  
11            return False  
12        i += 2  
13    return True  
14 num = int(input())  
15 print(is_prime(num))
```

 Run Save

Input

17

Output

True

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5 marks**Q6**

Fruitful Function (Returns One Value)



Q7 Function Returning Two or More Values

5 marks

Write a function that accepts a student's marks in three subjects and returns the total, average, and grade.

Your Code

```
1 def total_avg_grade(m1, m2, m3):  
2     total = m1 + m2 + m3  
3     avg = total / 3  
4  
5     # Simple grading based on percentage  
6     perc = avg  
7     if perc >= 90:  
8         grade = 'A'  
9     elif perc >= 80:  
10        grade = 'B'  
11    elif perc >= 70:  
12        grade = 'C'  
13    elif perc >= 60:  
14        grade = 'D'  
15    else:  
16        grade = 'F'  
17  
18    return total, avg, grade
```

Input

30 40 50

Output

120 40.0 F

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5 marks

Q6

Fruitful Function (Returns One Value)

Q8 Function Returning Two or More Values

5 marks

Write a function that accepts the radius of a circle and returns its area and circumference.

Your Code

```
1 import math
2
3 def circle_area_circumference(radius):
4     area = math.pi * radius * radius
5     circumference = 2 * math.pi * radius
6     return area, circumference
7 r = 5
8 a, c = circle_area_circumference(r)
9 print("Area:", a)
10 print("Circumference:", c)
```

 Run Save

Input

r

Output

```
Area: 78.53981633974483
Circumference: 31.41592653589793
```


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5 marks

Q6

Fruitful Function (Returns One Value)

Q9 Function Returning Two or More Values

5 marks

Write a function that accepts a number and returns its square, cube, and square root.

Your Code

```
1 import math
2
3 def square_cube_sqrt(n):
4     square = n ** 2
5     cube = n ** 3
6     sqrt = math.sqrt(n)
7     return square, cube, sqrt
8
9 # Example usage
10 x = float(input())
11 a, b, c = square_cube_sqrt(x)
12 print(a, b, c)
```

 Run Save**Input**

4

Output

16.0 64.0 2.0

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Fruitful Function (Returns One Value)

**Q10 Keyword Arguments**

5 marks

Write a function student(name, age, department) and call it using keyword arguments.

Your Code

```
1 def student(name, age, department):  
2     print("Name:", name)  
3     print("Age:", age)  
4     print("Department:", department)  
5  
6 student(name="rehan", age=18, department="AIML")
```

Input

rehan
18
AIML

Output

```
Name: rehan  
Age: 18  
Department: AIML
```

 Run Save